

SECOND ANNUAL IN-HOUSE CONVENTION

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DEPARTMENT OF PSYCHOLOGY

DALHOUSIE UNIVERSITY

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ABSTRACTS

1. R. Adamec - Emotional Biasing, an Experiential Tuning of Limbic Circuits?

The early experiential determinants of adult emotive biasing are being investigated on a behavioural and neurophysiological level. This research is intended to demonstrate a causal relationship between the impact of social experience on limbic system substrates of emotional expression, and epileptic excitability of those substrates -- in order to provide a brain-behaviour model for normal and schizo-affective emotive development.

2. R. Sutherland - The Relation Between Forebrain Activity and the Rewarding Effect of Electrical Brain Stimulation.

Anaesthetized rats will be trained to alter sciatic nerve activity for reinforcing brain stimulation. In order to identify neurons that may underly the rewarding effect, evoked potentials are recorded in forebrain structures. The role of selected regions in generating the rewarding effect can be assessed by chemical modification of their activity.

3. R. Blanchard - Observing Behaviour and Short-Term Memory.

Pigeons are run on an observing response procedure in which rewarded and nonrewarded trials occur randomly. Observing responses during a trial produce stimuli signalling trial outcome. Pretrial stimuli signalling trial outcome in advance lengthen the latency to observe and decrease the rate of observing behaviour.

4. W. K. Honig - Effective Cues for Working Memory in the Pigeon.

Pigeons remembered a blue or a green stimulus at the beginning of each trial in order to make an appropriate response controlling the end of the trial. Pecking at blue was reinforced, but not at green. Transfer tests showed that neither the occurrence of reinforcement nor differential pecking at the two colors was necessary to maintain working memory, but their combined absence led to a decrement in performance.

5. J. A. McNulty - Human Hearing Underwater: A few more facts and speculations.

Previous research done on the project has shown that divers can utilize their auditory sense to orient themselves and to navigate effectively underwater to a much greater extent than anyone had thought possible. In fact, man's ability to engage in these activities is almost as precise as it is in air. Such findings were surprising given the transformations imposed upon binaural cues by the increased speed of sound in water and the changes in acoustical impedance at the head-water interface. The similarity between results obtained in the two media suggest that the same underlying processes might be involved, contrary to what some have suggested. Results will be presented of some further work on auditory phenomena comparing performances in air and water which was done as preliminary to more intensive investigations of the auditory processes underlying these abilities.

6. F. Giffin and D. Mitchell - The Kitten Jumping Stand.

An adaptation of the Lashley jumping stand that enables rapid and sensitive estimates of visual acuity to be made in kittens as young as 4½ weeks of age will be demonstrated on film. Some applications of this technique will be described.

7. M. G. Yoon - Induction of the Field Compression within a Rotated Tectal Reimplant in Goldfish.

It was possible to induce the compression of visual projections onto a rotated tectal reimplant within the halved tectum. The field compression within the reimplanted area occurred along the original topographic polarity of the rotated tectal tissue. Thus the retention of polarity and readjustment of size-disparity are mutually compatible in the goldfish visual pathways.

8. G. Moran - Social Interactions in a Group of Wolves: An Interim Progress Report.

Detailed verbal descriptions of all aspects of observed social interactions among a group of eight wolves have been recorded regularly since October 1974. Representative interactions have been collected on high-speed film. Both classes of data are still accumulating and are in the process of being reduced and analysed. Although rigorous conclusions would be premature, some aspects of the pack's social organisation and the detailed structure of specific social interactions are becoming less unclear.

9. E. Renner - A Blueprint for Community Oriented Clinical/Social Psychology.

Society with increasing frequency will be confronted with ethical dilemmas arising from possible applications and interventions based upon new information from psychology as a behavioral and biological science. In a Ph.D. (as opposed to a professional or applied) program, clinical/social psychology has two roles: (a) to establish real world conceptual and theoretical constraints for focussing laboratory research, and (b) to articulate a philosophy of science which includes social responsibility and which is adequate for the emergent aspects of behavioural and biological psychology. A blueprint will be presented for how community oriented clinical/social psychology fills these roles and how such a program fits within a research department and within a university context.

10. W. J. Jacobs and V. M. LoLordo - Stimulus Control in Aversive Situations.

Rats were trained to press a bar to avoid electric shock and a) terminate a compound audio-visual warning stimulus or b) produce a compound audio-visual feedback stimulus. Stimulus control tests reveal the auditory element controls responding in a) and the visual element controls in b). The results are extended to the Pavlovian paradigm.

11. R. Klein - Boundary Conditions for the Object Superiority Effect.

Weisstein and Harris (1974) found that identification of a target line improves with the meaningfulness (3-dimensionality) of the pattern in which the line is embedded. This "object superiority effect" disappears when a random line mask, or speeded responses are used. Possible implications for perception will be discussed.

12. D. Mitchell and F. Giffin - The Influence of Visual Experience on the Development of Visual Acuity in Kittens.

Kittens were either reared in darkness from birth or were visually deprived for periods of from two weeks to two months after they were 45 or 60 days old. All animals eventually developed normal adult levels of acuity after periods of visual exposure of from three days for animals with prior visual experience to three months for animals with no visual experience.

13. P. Anderson - Interocular Transfer of the Threshold Elevation Aftereffect in Normals and Stereoblinds.

Interocular transfer of the threshold elevation aftereffect that follows adaptation to drifting gratings was investigated in normal and stereoblind subjects. Stereoblind subjects showed small amounts of transfer. In normals, as adapting grating contrast was increased, elevation increased in the adapted eye, but remained constant in the nonadapted eye. These data are considered in relation to currently held views of interocular transfer.

14. A. Randich - The Effect of a Conditioned fear stimulus on DRL Responding Depends upon the Context in which the DRL Schedule is Presented.

The rate of responding maintained by simple DRL schedules on the DRL component of a MULT DRL-EXT schedule decreased during a signal preceding shock. Conversely, the rate of responding maintained by the DRL component of a MULT DRL-VI schedule increased during a signal preceding shock.

15. J. Day - Right Hemisphere Word Recognition in Normal Right-Handed Adults.

Recent findings with commissurotomy patients have suggested that the disconnected right hemisphere, contrary to traditional views, has some capacity for processing language. A current question, however, is whether these findings reflect (a) language functions of the normal intact right hemisphere or (b) a functional anomaly resulting from a release of the right hemisphere from an inhibitory influence normally exerted by the left hemisphere. The present research provides data from normal subjects which supports the generality of the split-brain findings.

16. K. Bloom and C. Honig - Neonatal Behavioral Assessments.

The use of neurological and behavioural exams as methods for determining the sensory and response capacities of the human newborn will be discussed. We will explore how maternity hospitals provide a natural experimental setting for studying neonatal behaviours which influence, and are influenced by, the early social environment.

17. C. Adamec - Marihuana and men: Or Although on marihuana men don't have as much fun as women and are not as pleasant, they nevertheless do SOME interesting things.

18. G. Goddard, P. Nau and I. Golani - Strategies of Exploratory Behaviour in the Fornix-cut Rat.

P. Nau and I. Golani observe an interesting escalation in each object- or place-specific act during an exploratory sequence. O'Keefe and Nadel propose that the function of the hippocampus is to build cognitive maps of the environment. Disconnection of the hippocampus by fornix section, should disrupt all place-specific orderly progressions.

19. L. Davis - Analysis of Behavioural Sequences in a Damselfish.

An approach toward a more comprehensive quantitative description of an ethogram is taken with the pomacentrid, Eupomacentrus partitus. Emphasis is on the temporal patterning of motor, sonic and chromatic behaviour in males observed in both sea and aquarium.

20. B. Rusak - Having a Ball.

The cessation of reproductive activity during unfavourable seasons of the year is a vital ecological adaptation for most temperate zone species. In some rodents this seasonal reproductive cycle has been shown to be under the control of the pineal gland. Recent findings have demonstrated a crucial role for the suprachiasmatic nuclei (SCN) in the control of the pineal's activity. Planned studies will assess several alternative hypotheses as to the nature of SCN-pineal interaction and the mechanism whereby light influences this interaction.

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21. J. Willner - Effects of Prior Experience upon Subsequent Taste Aversion.

In comparison with rats poisoned five hours post-ingestion, rats poisoned immediately after drinking water in a novel environment reduce their consumption of fluid in that environment. The rats were then given saccharin in the novel environment, re-poisoned, and the effects of the chamber aversion on the taste aversion were assessed.

22. J. Gardner - Is the Cerebellum Indispensable?

Recovery from cerebellar damage is likely to be more complete if the lesion occurs early in life. The project I'll describe is aimed at assessing the effects of neonatal cerebellectomy on the cat oculomotor system through examination of spontaneous eye movements and the response and plasticity of the vestibulo-ocular reflex.

23. J. Fentress - Mouse Movements and Motives: Toward a Dynamic Synthesis.

The literatures on movement and motivation are traditionally separate. Yet the various motivational contexts in which particular movement patterns occur can provide important clues about motor control. My thesis is that both movement and motivational control systems exhibit parallel functions of interaction-self-organization. This suggests steps toward dynamic synthesis.

24. T. Barrett - Police-Youth Relations: A Social-Psychological Analysis.

Much of the rising crime rate across the country has been attributed to youth. This has been accompanied by increasing general hostility and disrespect towards the police by youth. Because the police lack open channels of communication with the public, they are one of the most isolated occupations, yet they must respond to the secondary by-products of social issues over which they have no control. The youths themselves represent an important aspect of social change. An analytical approach for establishing channels of communication between youth and police as well as future avenues for research directed towards both the police as a system and the youth as a subgroup will be discussed.

25. V.M. LoLordo, W. J. Jacobs and A.L. Riley - Proactive and Retroactive Effects of Physical Dependence on Taste Aversion Learning.

The development of physical dependence upon morphine, i.e. addiction to morphine, had no retroactive effect upon the aversion to a taste which had previously been paired with morphine, but did have a proactive, decremental effect upon formation of an aversion to a new taste paired with morphine.

26. D. M. Regan - Impaired Visual Function in Multiple Sclerosis Patients.

We have found that the temporal properties of visual perception are deranged in the demyelinating disease multiple sclerosis, even when standard tests reveal no abnormality of visual function. These findings can aid diagnosis, promise a means of monitoring the progress of demyelination and can give insights into normal visual function.

27. P. J. Urcuioli and W. K. Honig - The Effects of Prior Discrimination Training on the Acquisition of Matching-to-sample Performance in Pigeons.

Pigeons given prior training to discriminate between the stimuli subsequently used in a matching-to-sample problem learn the matching-to-sample task at a rate faster than pigeons given no such pretraining. If the prior discrimination is then "reversed", reacquisition of matching-to-sample performance appears retarded.

28. M. Cynader - Competition Among Sense Modalities in Development.

Single cells in the deeper layers of the cat superior colliculus can normally be activated by visual, auditory, or tactile stimuli. If kittens are dark-reared until four months of age, visual stimuli become ineffective. This, apparently permanent change in responsivity implies an experience-dependent competition among the different inputs for control of multimodal cells.

29. I. Golani (In collaboration with Philip Teitelbaum, J. Fentress, and P. Nau) The Organization of Exploratory Behaviour in the Rat.

Exploratory behaviour in the rat involves a gradual expansion of the amplitude of snout trajectories in Cartesian space. At first, only short amplitude horizontal scans are performed. As the amplitude of the scans increases, vertical scans are added. Still later, snout movements in the vertical, away from surfaces, appear. As the amplitudes of the snout movements increase, the various body segments are recruited in a regular manner. These rules apply to normal as well as to animals recovering from lateral hypothalamic lesions.

30. P. Jusczyk - When is a bag bad?: The perception of syllable final contrasts by two-month-olds.

Previous research on speech perception by infants has focussed on their ability to distinguish syllables differing in their initial segments (e.g. "ba" vs. "pa"). However, investigations of speech perception in adults have indicated that the cues used to distinguish the same phonetic contrast (e.g. "b" vs. "p") differ depending on the position of the contrast within a syllable. So, for example, while a voice-onset time cue is used to distinguish "ba" from "pa", vowel length is used to distinguish "ab" from "ap". Presently we are exploring the two-month-old's ability to distinguish syllables differing in their final positions.

31. B. McNaughton - Interactions between two Afferents to the Fascia Dentata: Evidence for Feedforward Inhibition.

Current literature maintains that the commissural afferents to the fascia dentata make excitatory contacts with granule cells. Contrary evidence will be presented, suggesting that the commissural fibers actually excite basket cell interneurons which, in turn inhibit the granule cells. The commissural afferents thus provide a powerful, feedforward, inhibitory influence on granule cell output.

32. R. Roseburgh and R. S. Rodger - Generating Right Linear Grammars.

The computer program used to find a right-linear grammar from the analysis of sequences of discrete events (behavioural acts) which have natural sequence termini has now been developed further. This talk will describe the sort of data suitable for such analysis, the role of grammars in such analysis and the role of "phrasing" and "equivalencing" in such grammars.

33. R. Douglas - Synaptic Potentiation in Hippocampal Slices.

Thin slices of hippocampus were maintained in an artificial environment for several hours. Each slice contained the major elements of the hippocampal circuit. Relatively long lasting potentiation was observed with stimulation of the perforant path and the Schaeffer collaterals in both the rat and hamster.

34. P. Nau - Behavioural Organization In Voles.

Two meadow voles have been housed in a large plexiglass environment capable of recording certain aspects of their behaviour continuously throughout the day. These records, along with detailed observational records, are being analyzed in hope that a description of the behavioural organization of these animals will emerge.

35. S. Nakajima - The Amnesic Effect of ECS Reduced by Hydrocortisone.

Injection of hydrocortisone into the ^{hippocampus} ~~rat~~ prior to a passive avoidance training reduces the amnesic effect of ECS given shortly after the training. It is probable that ECS produces amnesia by preventing corticosteroids from entering into the hippocampus to modify the neural activity in the limbic system.

36. D. Boness - Suckling Pattern and Related Behavior of Female and Pup Grey Seals, *Halichoerus grypus* at three breeding sites.

Major differences in caretaking behaviour have been suggested by ambiguous qualitative studies at different breeding sites. In this study, quantitative data concerning duration and frequency of suckling, amount of time females spend with their pups, and behavior toward unrelated pups were obtained at three separate breeding sites. The mean percentage of observation time females spent on land near their pups was 50%, 79% and 100%. Differences were also found in the mean duration of and mean frequency of suckling between mother and pup pairs. A high frequency of alien suckling occurred at one beach (75% of all females), but was relatively low at the other two breeding sites (20% and 6% of females). Possible explanations for these differences are discussed in terms of behavioral differences which might be mediated by the topography of the breeding site.

37. Z. Havkin - Body Contact in the Play Behaviour of Wolf Pups.

Body contact in wolf pups is analyzed by means of a fine grain method. It is suggested that such an analysis is necessary for the understanding of some aspects of play behaviour. Some behavioural regularities extracted from analysed sequences are discussed.

38. J. Barresi - Set-theoretic Approaches to Quantitative Semantics

Quantitative approaches to semantics based on the theory of mathematical sets will be described and relevant data reported. Both traditional set theory and the more recent fuzzy set theory might prove useful as models of semantic processing. In the present paper, data is reported which indicates the fruitfulness of both these approaches.

39. R. S. Rodger - The Machinations of Manova.

Somewhat like Mata Hari, Multivariate Analysis of Variance has many mysteries and magical properties. This talk will discuss some of these; in particular the three ways of judging what is happening, whether the questions about what is happening are relevant, and when judgments are better on the vertical (standing up) or horizontal (lying down) dimensions.

40. F. H. Harrington - Vocal Behaviour in Wolves

An individual's behavior is influenced by its social context. During a group howling session, a wolf's howls evidently reflect this influence through their structure. Using film and sonographic analysis, howling interactions in red wolf-coyote hybrids at Shubenacadie are being studied to describe howl structure and its possible relation to social factors.

41. T. Whalen - Matching on Mixed Concurrent and Multiple Schedules.

When the discriminative stimuli are removed from multiple and concurrent schedules of reinforcement, if pigeons on both types of schedules distribute their responses between the two component schedules in the same proportion as the reinforcements are distributed it can be argued that the subject's responses to change in the component schedules are determined by the interval that has passed since the last reinforcement, whereas, if only the subjects on the concurrent schedules distribute their responses in this manner, then it can be argued that the subject uses the probabilities that responses are followed by reinforcement and that changes in the component schedule followed by responses are followed by reinforcement determine when the subject changes component schedules.